Status	Finished
Started	Thursday, 30 October 2025, 12:46 PM
Completed	Thursday, 30 October 2025, 1:04 PM
Duration	18 mins 38 secs
Marks	1.00/3.00
Grade	**1.00** out of 3.00 (**33.36**%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade**.

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question **1**

Incorrect

Mark 0.00 out of 1.00

The following sequence

$$p_n = (-2)^{n+1}$$

is a particular solution to the recurrence relation

$$\frac{a_n}{2} + 2 \cdot a_{n-1} = (-2)^n \, .$$

a) Find another solution to the recurrence relation, not equal to p_n:

$a_n =$ 1/a[n-1]

Your last answer was interpreted as follows:

$$\frac{1}{a_{n-1}}$$

The variables found in your answer were: $[a_{n-1}]$

Syntax guide: Your answer should be an expression which depends on the variable n, and *no other constants*. E.g. if you think the answer is

$$-7 \cdot \frac{2^n + 1}{3n} \, ,$$

you should type **-7*(2^n+1) / (3*n)**.

b) Given the initial condition $a_0 = -3$. Compute a_n for $n = 1$:

$a_1 =$ 1/-3

Your last answer was interpreted as follows:

$$\frac{1}{-3}$$

Question **2**

Partially correct

Mark 0.50 out of 1.00

In this problem you are tasked with computing with elements in the symmetric group of permutations, $(S_5, \cdot)$.

Syntax guide: The answer to each task is an element of the group S_5 and should be given as a 2×5 matrix. E.g. if you think the answer to a question is the permutation denoted by

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix}$$

then you should answer

```
1  2  3  4  5
2  3  1  5  4
```

a)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 2 & 4 & 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 1 & 5 \end{bmatrix} =$$

```
1 2 3 4 5
5 3 1 4 2
```

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 3 & 1 & 4 & 2 \end{bmatrix}$$

b)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 1 & 5 & 4 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 5 & 2 & 3 & 4 \end{bmatrix} =$$

```
1 2 3 4 5
2 1 4 3 5
```

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 4 & 3 & 5 \end{bmatrix}$$

c)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 2 & 4 & 3 & 1 \end{bmatrix}^{-1} =$$

```
5 2 4 3 1
1 2 3 4 5
```

Your last answer was interpreted as follows:

$$\begin{bmatrix} 5 & 2 & 4 & 3 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix}$$

d)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 1 & 5 \end{bmatrix}^{-1} =$$

```
2 3 4 1 5
1 2 3 4 5
```

Your last answer was interpreted as follows:

$$\begin{bmatrix} 2 & 3 & 4 & 1 & 5 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix}$$

Question **3**

Partially correct

Mark 0.50 out of 1.00

a) Let $G = (\mathbb{Z}/2)^{\times 4}$, and let $H \subset G$ be the subgroup

$$H = \{[0,0,0,0]\,,[0,0,1,0]\,,[0,1,0,1]\,,[0,1,1,1]\,,[1,0,0,1]\,,[1,0,1,1]\,,[1,1,0,0]\,,[1,1,1,0]\}\,.$$

What is the index of $H \subset G$?

$[G : H]$ = 2

Your last answer was interpreted as follows:

$$2$$

b) Let D_4 be the dihedral group of order 8 of symmetries of the unit square centered at the origin of $\mathbb{R}^2$. Let $\sigma \in D_4$ be the symmetry that rotates 90 degrees around the origin.

Compute the index of the subgroup generate by σ:

$[D_4 : \langle\sigma\rangle]$ = 4

Your last answer was interpreted as follows:

$$4$$

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 305

Signing code: 20525

Your last answer was interpreted as follows:

$$20525$$

◄ Test 2 (topics 4-6: Languages, Machines, Regular expressions, Correct programs, Relations)

Jump to...